

Experiment ‘p_6_2_a_swap_first_last_seq’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_6_2_a_swap_first_last_seq

Natural language statement: Write a method to swap the first and last elements in an array.

Method signature: p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `result[0] == inputs[|inputs| - 1]`
- `result[|inputs| - 1] == inputs[0]`
- `forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]`

Requires

- `|inputs| >= 2`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to swap the first and last elements in an array.

The signature should be:

`p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)`

The method should respect the following contract:

`requires |inputs| >= 2, ensures |result| == |inputs|, ensures result[0] == inputs[|inputs| - 1], ensures result[|inputs| - 1] == inputs[0], ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0B0Uhq0nsq3VLIUaVPAwii2PoOc

Model: gpt-5.5-2026-04-23

Created at: 1780571122

Finish reason: stop

Usage: 390 tokens in, and 4951 tokens out

```
{
  var n := |inputs|;

  assert 0 <= 0 < |inputs|;
  assert 0 <= n - 1 < |inputs|;
  var tmp := inputs[0 := inputs[n - 1]];
  assert |tmp| == n;

  assert 0 <= n - 1 < |tmp|;
  result := tmp[n - 1 := inputs[0]];
  assert |result| == n;

  assert 1 <= n - 1;
  assert n - 1 != 0;
  assert 0 <= 0 < |result|;
  assert 0 <= 0 < |tmp|;
  assert result[0] == tmp[0];
  assert tmp[0] == inputs[n - 1];
  assert result[0] == inputs[n - 1];
}
```

```

assert 0 <= n - 1 < |result|;
assert result[n - 1] == inputs[0];

forall i | 1 <= i < n - 1
  ensures result[i] == inputs[i]
{
  assert 0 <= i;
  assert i < n;
  assert i < |result|;
  assert i < |tmp|;
  assert i < |inputs|;
  assert i != n - 1;
  assert i != 0;
  assert result[i] == tmp[i];
  assert tmp[i] == inputs[i];
}
}

```

Final Program

```

method p_6_2_a_swap_first_last_seq(inputs: seq<int>) returns (result: seq<int>)
  requires |inputs| >= 2
  ensures |result| == |inputs|
  ensures result[0] == inputs[|inputs| - 1]
  ensures result[|inputs| - 1] == inputs[0]
  ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == inputs[i]
{
  var n := |inputs|;

  assert 0 <= 0 < |inputs|;
  assert 0 <= n - 1 < |inputs|;
  var tmp := inputs[0 := inputs[n - 1]];
  assert |tmp| == n;

  assert 0 <= n - 1 < |tmp|;
  result := tmp[n - 1 := inputs[0]];
  assert |result| == n;

  assert 1 <= n - 1;
  assert n - 1 != 0;
  assert 0 <= 0 < |result|;
  assert 0 <= 0 < |tmp|;
  assert result[0] == tmp[0];
  assert tmp[0] == inputs[n - 1];
  assert result[0] == inputs[n - 1];

  assert 0 <= n - 1 < |result|;
  assert result[n - 1] == inputs[0];
}

```

```
forall i | 1 <= i < n - 1
  ensures result[i] == inputs[i]
{
  assert 0 <= i;
  assert i < n;
  assert i < |result|;
  assert i < |tmp|;
  assert i < |inputs|;
  assert i != n - 1;
  assert i != 0;
  assert result[i] == tmp[i];
  assert tmp[i] == inputs[i];
}
}
```

Total Token Usage

Input tokens: 390

Output tokens: 4951

Reasoning tokens: 4542

Sum of 'total tokens': 5341

Experiment Timings

Overall Experiment started at 1780571122631, ended at 1780571232803, lasting 110172ms (110.17 seconds)

Iteration #1 started at 1780571122631, ended at 1780571232803, lasting 110172ms (110.17 seconds)